# Protein Sequence Analysis of Cytochrome c Using BLAST

## OBJECTIVE:

The objective of this task is to retrieve the protein sequence of Cytochrome c from Homo sapiens using the NCBI database and perform sequence similarity analysis using the BLAST tool. The study aims to identify homologous sequences, determine percentage identity, analyze query coverage, evaluate E-value significance, and understand evolutionary conservation of the protein across different species.

## INTRODUCTION:

Bioinformatics is an interdisciplinary field that combines biology, computer science, and data analysis to study biological data. One of the important applications of bioinformatics is protein sequence analysis, which helps in understanding the structure, function, and evolutionary relationships of proteins.

The Basic Local Alignment Search Tool (BLAST) is a widely used tool for comparing a query sequence with sequences available in biological databases. It helps identify similar or homologous sequences based on parameters such as percentage identity, query coverage, alignment score, and E-value. These parameters indicate the level of similarity and the statistical significance of the matches.

Cytochrome c is a small heme protein that plays a crucial role in the electron transport chain and cellular respiration. Due to its essential biological function, this protein is highly conserved across different species. In this study, the protein sequence of Cytochrome c from Homo sapiens was retrieved from the NCBI database and analyzed using BLAST to identify homologous sequences and study its evolutionary conservation.

**METHODOLOGY:**

1. **Sequence Retrieval**

   The protein sequence of Cytochrome c from *Homo sapiens* was retrieved from the National Center for Biotechnology Information (NCBI) protein database. The sequence was downloaded in FASTA format.

2. **BLAST Analysis**

   The retrieved FASTA sequence was submitted to the NCBI BLAST (Basic Local Alignment Search Tool) using the BLASTP program for protein sequence comparison.

3. **Database Selection**

   The search was performed against the non-redundant (nr) protein database using default parameters to identify similar protein sequences across different organisms.

4. **Result Analysis**

   The BLAST output was analyzed to identify homologous sequences. Important parameters such as query coverage, percentage identity, alignment score, and E-value were recorded.

5. **Alignment Examination**

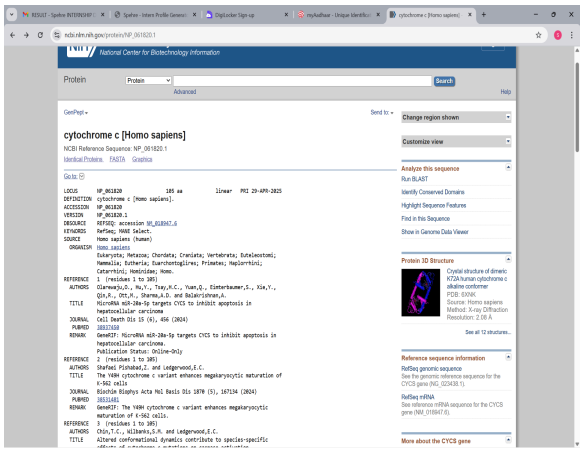
   The top matching sequences were examined in the alignment section to observe conserved regions and amino acid similarities.

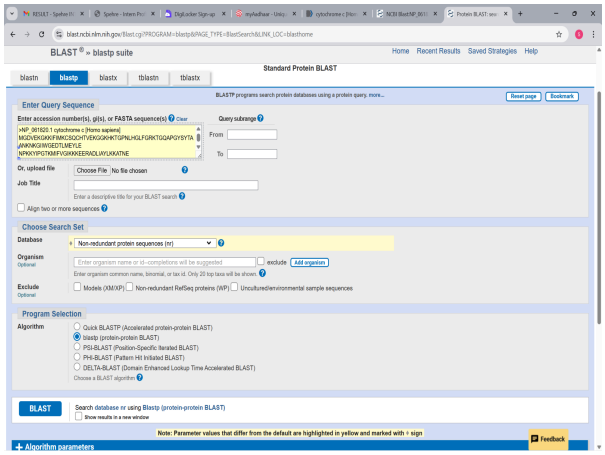
## RESULTS:

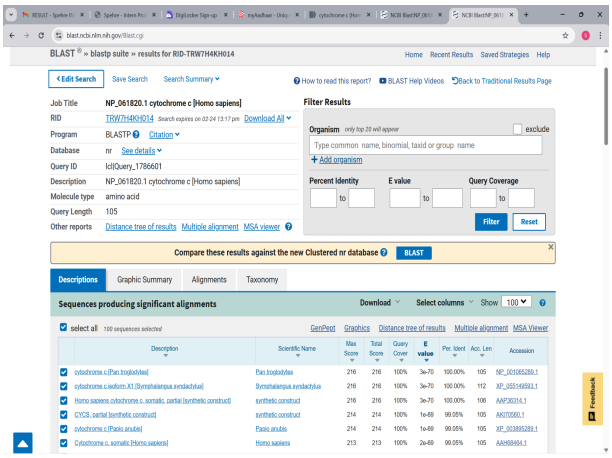
The protein sequence of Cytochrome c from *Homo sapiens* was analyzed using the BLASTP program against the NCBI non-redundant (nr) protein database. The BLAST search returned several significant matches from different species.The alignment score for the top match was 216 bits, with an E-value of 3e-70, which indicates a highly significant similarity and confirms that the match is not due to random chance.The graphical summary showed full-length alignment of the query sequence with most of the subject sequences. The conserved domain Cyc7, which belongs to the cytochrome c family, was also detected, confirming the functional classification of the protein.

## CONCLUSION:

The BLAST-based protein sequence analysis of Cytochrome c from *Homo sapiens* demonstrated a high degree of similarity with homologous sequences from related species, particularly primates, showing 100% query coverage and very high percentage identity with extremely low E-values. The alignment results confirmed strong evolutionary conservation with no significant gaps or variations in closely related organisms. These findings indicate that Cytochrome c is a highly conserved and functionally essential protein, reflecting its critical role in cellular respiration and energy production across species.





blast.ncbi.nlm.nih.gov/Blast.cgi

100 sequences selected

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**cytochrome c [Pan troglodytes]**

Sequence ID: NP_001065289.1 Length: 105 Number of Matches: 1

See 24 more title(s) See all Identical Proteins(IPG)

Range 1: 1 to 105 GenPept Graphics ▼Next Match ▲Previous Match

| Score | Expect | Method | Identities | Positives | Gaps |
|---|---|---|---|---|---|
| 216 bits(549) | 3e-70 | Compositional matrix adjust. | 105/105(100%) | 105/105(100%) | 0/105(0%) |

```
Query  1    MGDVEKGKKIFIMKCSQCHTVEKGGKHKTGPNLHGLFGRKTGQAPGYSYTAANKNKGIIW  60
            MGDVEKGKKIFIMKCSQCHTVEKGGKHKTGPNLHGLFGRKTGQAPGYSYTAANKNKGIIW
Sbjct  1    MGDVEKGKKIFIMKCSQCHTVEKGGKHKTGPNLHGLFGRKTGQAPGYSYTAANKNKGIIW  60

Query  61   GEDTLMEYLENPKKYIPGTKMIFVGIKKKEERADLIAYLKKATNE  105
            GEDTLMEYLENPKKYIPGTKMIFVGIKKKEERADLIAYLKKATNE
Sbjct  61   GEDTLMEYLENPKKYIPGTKMIFVGIKKKEERADLIAYLKKATNE  105
```

**Related Information**
Gene - associated gene details
AlphaFold Structure - 3D structure displays
Genome Data Viewer - aligned genomic context
Identical Proteins - Identical proteins to NP_001065289.1

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**cytochrome c isoform X1 [Symphalangus syndactylus]**

Sequence ID: XP_055149593.1 Length: 112 Number of Matches: 1

See 1 more title(s) See all Identical Proteins(IPG)

Range 1: 8 to 112 GenPept Graphics ▼Next Match ▲Previous Match

| Score | Expect | Method | Identities | Positives | Gaps |
|---|---|---|---|---|---|
| 216 bits(550) | 3e-70 | Compositional matrix adjust. | 105/105(100%) | 105/105(100%) | 0/105(0%) |

```
Query  1    MGDVEKGKKIFIMKCSQCHTVEKGGKHKTGPNLHGLFGRKTGQAPGYSYTAANKNKGIIW  60
            MGDVEKGKKIFIMKCSQCHTVEKGGKHKTGPNLHGLFGRKTGQAPGYSYTAANKNKGIIW
Sbjct  8    MGDVEKGKKIFIMKCSQCHTVEKGGKHKTGPNLHGLFGRKTGQAPGYSYTAANKNKGIIW  67

Query  61   GEDTLMEYLENPKKYIPGTKMIFVGIKKKEERADLIAYLKKATNE  105
            GEDTLMEYLENPKKYIPGTKMIFVGIKKKEERADLIAYLKKATNE
Sbjct  68   GEDTLMEYLENPKKYIPGTKMIFVGIKKKEERADLIAYLKKATNE  112
```

**Related Information**
Genome Data Viewer - aligned genomic context
Identical Proteins - Identical proteins to XP_055149593.1

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**Homo sapiens cytochrome c, somatic, partial [synthetic construct]**

blast.ncbi.nlm.nih.gov/Blast.cgi

**CYCS, partial [synthetic construct]**

Sequence ID: AKI70560.1 Length: 105 Number of Matches: 1

Range 1: 1 to 105 GenPept Graphics ▼Next Match ▲Previous Match

| Score | Expect | Method | Identities | Positives | Gaps |
|---|---|---|---|---|---|
| 214 bits(545) | 1e-69 | Compositional matrix adjust. | 104/105(99%) | 104/105(99%) | 0/105(0%) |

```
Query  1    MGDVEKGKKIFIMKCSQCHTVEKGGKHKTGPNLHGLFGRKTGQAPGYSYTAANKNKGIIW  60
            MGDVEKGKKIFIMKCSQCHTVEKGGKHKTGPNLHGLFGRKT QAPGYSYTAANKNKGIIW
Sbjct  1    MGDVEKGKKIFIMKCSQCHTVEKGGKHKTGPNLHGLFGRKTSQAPGYSYTAANKNKGIIW  60

Query  61   GEDTLMEYLENPKKYIPGTKMIFVGIKKKEERADLIAYLKKATNE  105
            GEDTLMEYLENPKKYIPGTKMIFVGIKKKEERADLIAYLKKATNE
Sbjct  61   GEDTLMEYLENPKKYIPGTKMIFVGIKKKEERADLIAYLKKATNE  105
```

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**cytochrome c [Papio anubis]**

Sequence ID: XP_003895289.1 Length: 105 Number of Matches: 1

See 42 more title(s) See all Identical Proteins(IPG)

Range 1: 1 to 105 GenPept Graphics ▼Next Match ▲Previous Match

| Score | Expect | Method | Identities | Positives | Gaps |
|---|---|---|---|---|---|
| 214 bits(544) | 1e-69 | Compositional matrix adjust. | 104/105(99%) | 104/105(99%) | 0/105(0%) |

```
Query  1    MGDVEKGKKIFIMKCSQCHTVEKGGKHKTGPNLHGLFGRKTGQAPGYSYTAANKNKGIIW  60
            MGDVEKGKKIFIMKCSQCHTVEKGGKHKTGPNLHGLFGRKTGQAPGYSYTAANKNKGI W
Sbjct  1    MGDVEKGKKIFIMKCSQCHTVEKGGKHKTGPNLHGLFGRKTGQAPGYSYTAANKNKGIIW  60

Query  61   GEDTLMEYLENPKKYIPGTKMIFVGIKKKEERADLIAYLKKATNE  105
            GEDTLMEYLENPKKYIPGTKMIFVGIKKKEERADLIAYLKKATNE
Sbjct  61   GEDTLMEYLENPKKYIPGTKMIFVGIKKKEERADLIAYLKKATNE  105
```

**Related Information**
Gene - associated gene details
AlphaFold Structure - 3D structure displays
Genome Data Viewer - aligned genomic context
Identical Proteins - Identical proteins to XP_003895289.1

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**Cytochrome c, somatic [Homo sapiens]**

Sequence ID: AAH68464.1 Length: 105 Number of Matches: 1

Range 1: 1 to 105 GenPept Graphics ▼Next Match ▲Previous Match

| Score | Expect | Method | Identities | Positives | Gaps |
|---|---|---|---|---|---|

**Related Information**
Gene - associated gene details

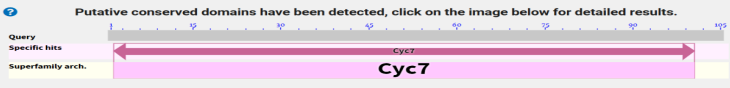
blast.ncbi.nlm.nih.gov/Blast.cgi

Descriptions | **Graphic Summary** | Alignments | Taxonomy

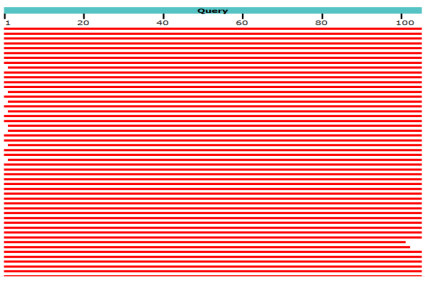
hover to see the title | click to show alignments | Show Conserved Domains | Alignment Scores: < 40 | 40 - 50 | 50 - 80 | 80 - 200 | >= 200

100 sequences selected

**Putative conserved domains have been detected, click on the image below for detailed results.**




**Distribution of the top 100 Blast Hits on 100 subject sequences**




https://www.ncbi.nlm.nih.gov/Structure/cdd/wrpsb.cgi?RID=TRW7HXM1016&FULL